# Chamod Shyamal Madusan

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Chamod Shyamal

## Education

**•University of Moratuwa** — Sri Lanka
B.Sc. (Hons) in Electronic and Telecommunication Engineering — 2021 – 2025
–**GPA: 3.72/4.00** ( Up to Semester 7 )

**•Richmond College, Galle** — Sri Lanka
G.C.E. Advanced Level Examination 2019 (Physical Science Stream) — 2006 – 2019
–Ranked: $50^{th}$ in the country and $5^{th}$ in the district with a z-score of 2.7754 (Top 0.28%)

**Trainee Full Stack Developer - University of Moratuwa (Open Learning Platform)**
- Python for Beginners
- Python Programming
- Server-side Web Programming
- Web Design for Beginners
- Front-End Web Development
- Professional Practice in Software Development

**Other Courses**
- Deep Learning Specialization - *DeepLearning.AI (Coursera)*
- Game Design and Development with Unity 2020 Specialization - *Michigan State University (Coursera)*
- Generative AI with Large Language Models - *DeepLearning.AI , Amazon Web Services (Coursera)*

## Projects

**• Enfluent - language learning platform (enfluent.skillsurf.lk)** — 2024-25
Developed a personalized language learning platform.
– Focuses on learning reading, writing, and speaking based on IELTS criteria.
– Provides individual AI grading for each criterion to evaluate progress.
– Created a 3D AI chatbot avatar provides real-time feedback on speech, including grammar, pronunciation, and confidence levels, enhancing learner engagement.
– Incorporated data from embedded systems (e.g., heart rate and emotional analysis) to adapt lesson plans and provide tailored feedback for individual learners.
– Tools & technologies used: *React , Deepface , OpenAI , Python , ThreeJS , Blender , JavaScript , Rhubarb , Java*

**• Cyborg Fury - Third Person Shooting game Design & Develop** — 2024-25
Create an RPG shooting game with realistic 3D graphics using the Unreal game engine.
– Developed core mechanics like third-/first-person mode switching, weapon pickups, dodging, and tactical movement (ladder, elevator, cover).
– Includes intelligent enemy AI and customizable graphics for enhanced game-play
– Designed intelligent enemy AI with a player-buddy system for cooperative gameplay.
– Added vehicle control for spaceships and sci-fi vehicles to enhance exploration and combat.
– Implemented melee finishers, drones, turrets, power-ups, and sniper functionality for combat variety.
– Created a customizable UI with graphics settings and a fully functional main menu.
– Built with Unreal Engine 5.4 and Blueprints, the game offers immersive realistic graphics.

**• Open world Game Implementation Using Unity with Questionnaire Web Application** — 2024
Start an exciting 3D RPG adventure in a open world , made for Unity WebGL platform.
– Beautifully optimized worlds and an engaging narrative.
– Includes an open-world adventure with realistic mechanics and quests like shooting, hunting, farming and stealing.
– Includes advanced weapons, vehicle control, a shop system, dynamic weather, and realistic gameplay.
– Develop a questionnaire website focused on promoting energy-saving habits and sustainable practices.
– Tools & technologies used: *Unity Engine , Character Creator 4 , Iclone 8 , C# , Java , JavaScript , React*

**• Crypto Trading Platform - Backend Development** — 2024-25
Developed and tested backend services for a cryptocurrency trading platform.
– Implemented backend functionalities for buy & sell crypto transactions, portfolio management, advanced wallet operations.
– Developed secure wallet-to-wallet transfers, withdrawal to bank accounts, and wallet balance top-up APIs.
– Built robust authentication system with login, registration, two-factor authentication, and password recovery.
– Performed thorough API testing and validation using Postman to ensure reliability and security.
– Tools & technologies used: *Spring Boot, Stripe , MySQL, Spring Security, Java Mail Sender, Postman, Java*

**• Simple E-commerce Application Website Development (shay.store)** — 2024
Developed a comprehensive e-commerce website tailored for selling headphones, speakers, and smartwatches.
– Integrated payment processing, a user-friendly shopping cart, and efficient item management for seamless e-commerce functionality.
– Built with Next.js and React for a responsive, visually engaging, and intuitive shopping experience, powered by Sanity for backend operations.
– Tools & technologies used: *React , NextJS , Sanity , JavaScript , CSS , Stripe*

**• Securing a Smart Building System** — 2025
Conducted multi-phase security evaluation and defense strategy development for IoT-enabled smart buildings.
– Identified crypto and access control flaws in smart building; mapped to CVEs and proposed mitigations.
– Built and tested secure encryption, decryption, and hashing APIs using Python (Flask, Cryptography).
– Conducted penetration testing on exposed APIs; documented findings and recommended fixes.
– Implemented rule-based intrusion detection for login abuse, spamming, and sensor anomalies; validated via simulations.
– Tools: *Python, Flask, Cryptography, Postman, CVE Database, API Testing, Wireshark*

- **Ancient Language Character Translation** 2025

Developed a Sinhala ancient handwritten character recognition system using deep learning and ensemble models.
  - Designed a two-stage classification system for 88 ancient Sinhala characters and their historical eras using deep learning and traditional ML.
  - Applied transfer learning with MobileNetV2 and fine-tuned dense layers for era classification across 37 datasets.
  - Extracted deep features using VGG19, ResNet50, InceptionV3, and InceptionResNetV2, followed by ensemble classification.
  - Enhanced model robustness through advanced data augmentation including shearing, brightness shift, and zoom transformations.
  - Tools & technologies used: *Python, TensorFlow, Keras, Scikit-learn, MobileNetV2, OpenCV, XGBoost*

- **Brain Haemorrhage Detection** 2023

Developed a brain hemorrhage detection and localization system to identify intracranial hemorrhages.
  - Three models were analyzed, their performance metrics were defined, and ResNet150 was chosen for its effectiveness in identifying intracranial hemorrhages.
  - Six windowing methods were utilized, and the sigmoid BSB method was identified as the most effective.
  - Preprocessing data from a CSV file, comparing with the SOP Instance UID, and then opening the corresponding DICOM image and labels..
  - Tools & technologies used: *Python , Tensorflow , Numpy , Pandas*

- **Design of Local Area Network Simulation** 2023

Implemented a backbone network for the university and the ENTC department.
  - Configured the backbone network routing with OSPF and simulated it using Packet Tracer.

- **Mood Lamp (Individual)** 2023

Created a customizable mood lamp with remote control capabilities.
  - Features include remote color and pattern control via a mobile app, real-time brightness adjustments, and scheduled mood changes.
  - Tools & technologies used: *Arduino/C++, Altium, Solidworks*

- **Virtual and physical robot implementation**

Designed two distinct implementations, both physical and virtual, to perform specific tasks 2022 - 2023
  - wall following, line maze solving, finding the shortest return path, escaping from the blind box , dotted line following, segmented wall following, chess board activity, complete the broken bridge by placing boxes
  - Tools & technologies used: *Webots, SolidWorks, Arduino/C++*

## Internship Experience

- **MAS Holdings Bodyline Pvt Ltd - Machine Learning Engineer (Autonomation Team)**

- Developed a computer vision model for error detection in labels , utilizing Raspberry Pi for model deployment, and created a local web app to run the error detection model on Raspberry Pi.
- Optimize model for embedded-level deployment and accelerate using Google Coral TPU.
- Developed PLC programs for reconstructing and programming Automated Guided Vehicles (AGVs), emphasizing security and reliability.
- Other Skills: SolidWorks, Project Management, Electronics

## Technical Skills and Interests

**Programming Languages**:Python(intermediate), C++(basic), C# (intermediate with Unity), JavaScript(basic), MATLAB(basic) , Blueprint(Unreal Engine)

**Developer Tools**: HTML, CSS, Numpy, Pytorch, Tensorflow, React, ThreeJS, Microsoft Power Apps

**EDA, Simulation and CAD Tools**: Unity , Altium, Proteus, SolidWorks, Webots , Unreal Engine 5

**Languages**: : Sinhala (native) , English

## Volunteering and Leadership Experience

- **Committee Member ,** Electronic Club of University of Moratuwa 2022 - Present
- **Department Facilitator ,** EXMO23 The Flagship Technological Exhibition of University of Moratuwa 2023
- **Events Pillar ,** ACIES seson 02 Gaming Competition organized by Mora Esports Club 2022
- **Committee Member ,** Esports Club of University of Moratuwa 2021 - Present
- **Basket Ball Vice Captain ,** Richmond College Galle 2017-2018

## Academic Achievements

–**Mahapola Higher Education (Merit) Scholarship** 2020

*For outstanding performance in GCE A/L Examination.*

–**The Best All-Round Student Enter the University.** 2020

*Richmond College, Galle, Sri Lanka*

–**Physics Olympiad - Silver Medal** 2019

*Institute of Physics, Sri Lanka*

## References

**Dr. Ranga Rodrigo**
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